

Trajectory temperature, integrity, and quality specification

Stage: 11E-STAT0

First implementation: mdstats 0.20.19a0

Status: normative

Purpose

STAT0 converts the immutable ENS0 source evidence and ENS1 control certificate into an equally immutable source-segment quality assessment. It reconstructs ionic temperature from ionic kinetic energy, tests catastrophic integrity separately from manageable numerical degradation, evaluates the ensemble-appropriate conserved quantity when one is available, and produces one of exactly three execution verdicts:

`strictly_qualified`

`degraded_quality`

`unqualified`

Only `unqualified` blocks production scientific analysis by default. `degraded_quality` emits one warning and remains available to downstream analyses with all failed checks, effect sizes, assumptions, and source signatures attached.

STAT0 does not select a production interval, certify stationarity, or authorize a PMF. Those responsibilities remain in STAT1 and STAT2.

Inputs

The source-general evaluator consumes:

- one normalized `AtomisticFrameCollection` with trajectory semantics;
- one exact `FrameEnergyCatalog` aligned to the evaluated frames;
- one `NumericalMDQualityControls` record;
- one signed `SimulationControlCertificate`;
- the matching `SourceTrajectoryBundleIdentity` signature;
- one versioned `TrajectoryQualityPolicy`.

The source identity, control bundle, coordinate payload, and frame axis must describe the same source trajectory bundle. A user comment such as VASP SYSTEM is never an input to quality inference.

Ionic temperature

The instantaneous ionic temperature is

$$T_t = \frac{2K_{\text{ion}}(t)}{f_{\text{ion}}k_{\text{B}}}.$$

The `IonicTemperatureDefinition` records:

- the exact kinetic-energy channel and source path;
- atom count and active ionic coordinate count;
- fixed-coordinate and constraint evidence;
- whether center-of-mass translation is excluded;
- the resulting ionic degrees of freedom;
- any source-default or approximation note.

The first VASP implementation uses the documented periodic-system convention $3N - 3$ when all atoms are represented, the cell is periodic, and no affirmative fixed-coordinate count is available. The approximation is retained in metadata rather than hidden. Bound constraint evidence will supersede this default in later adapters.

`IonicTemperatureStatistics` retains:

- the complete derived temperature series;
- represented-time mean and standard deviation;
- minimum and maximum;
- integrated autocorrelation time and effective sample count;
- block length, block means, and block count;
- standard error and confidence interval of the mean;
- drift slope and confidence interval;
- a temperature-stability outcome.

The standard deviation is a physical fluctuation. Confidence in the mean and trend is computed from autocorrelation-aware complete-system blocks; adjacent frames are never counted as independent observations.

Integrity and numerical-quality checks

Each check is an immutable `TrajectoryQualityCheck` with:

```
check_id
severity = hard_integrity | soft_quality | diagnostic
requirement = hard_integrity_required | verdict_critical | method_specific | optional
status = pass | fail | warning | unavailable | insufficient | not_applicable
measured_value
threshold
units
message
evidence
```

Hard integrity

The first implementation checks at least:

- positive frame and atom counts;

- complete source positions and cells;
- finite normalized coordinates, cells, forces, and energy channels;
- positive nonsingular cell determinants;
- strictly increasing and uniformly spaced source steps and physical times;
- persistent atom identity through the normalized collection contract;
- no catastrophic periodic atomic overlap;
- no runaway force, speed, energy jump, or energy drift;
- no internal source-frame gaps.

A hard failure produces `unqualified` and `TrajectoryIntegrityError` unless diagnostic mode explicitly requests the certificate without continuation.

Soft quality

Soft checks include:

- electronic convergence reaching NELM;
- EDIFF relative to the policy target for precision MD;
- PREC, LREAL, ROPT, and timestep diagnostics;
- requested-versus-present ionic-step mismatch;
- temperature-mean confidence and temperature trend;
- ensemble-appropriate energy conservation;
- source energy-identity consistency.

A soft failure or material warning produces `degraded_quality`, emits exactly one `TrajectoryDegradedQualityWarning`, and does not block downstream execution. A genuinely optional and method-irrelevant missing channel is `unavailable` and does not degrade the trajectory.

Diagnostic requirement matrix

`DiagnosticRequirement` is the persistent classification for each check. Severity and requirement are independent metadata. Severity controls the top-level execution verdict; requirement states how a downstream method may use the evidence:

Requirement	Meaning	Missing or failed evidence
<code>hard_integrity_required</code>	Mandatory for any production trajectory analysis	Produces <code>unqualified</code> when the hard check fails
<code>verdict_critical</code>	Required for the general quality verdict	Produces <code>degraded_quality</code> when soft evidence fails or is materially insufficient

Requirement	Meaning	Missing or failed evidence
<code>method_specific</code>	Required only by a named downstream estimator	Does not change the general execution verdict; that method must fail closed or downgrade its own certificate
<code>optional</code>	Informative diagnostic only	Remains unavailable without degrading the trajectory

The requirement field prevents optional absent data from degrading an otherwise usable trajectory and prevents the general quality verdict from silently authorizing a method whose own required evidence is absent.

Realized ensemble consistency

STAT0 creates a signed `RealizedEnsembleConsistency` record separate from the ENS1 control-inferred ensemble. The first implementation checks the realized fixed-cell NVE behavior using:

- observed cell-volume range and full cell-matrix deviation;
- inactive Nosé-energy channels, when present;
- source-reported NVE total-energy drift.

Its status is one of:

`consistent`
`degraded`
`insufficient`
`inconsistent`

A degraded NVE consistency record contributes evidence to method-specific admissibility but does not redefine the ENS1 ensemble and does not by itself block structural analysis. The first implementation is explicitly diagnostic-only for non-NVE ensembles; source- specific extended-Hamiltonian checks belong to later adapters.

NVE energy conservation

For a resolved NVE source with a complete source-reported `total` channel, STAT0 records:

- initial, final, mean, and standard deviation;
- linear drift in eV/ps and eV/(atom ps);
- observation-span change;

- detrended residual standard deviation;
- maximum adjacent-frame energy jump;
- source identity residual, when `total`, ionic kinetic, and the relevant electronic energy channel are all available.

The current policy separates a strict conservation tolerance from a hard drift threshold. The hard value corresponds to approximately one room-temperature kT per atom over a 1 ps reference interval. A smooth, measurable drift may therefore yield `degraded_quality` without becoming `unqualified`.

For thermostat or variable-cell ensembles, STAT0 records available energy diagnostics but does not pretend that the NVE `total` channel is the conserved quantity. Extended Hamiltonian handling remains source- and ensemble-specific.

Default policy v2

The initial default policy is versioned and serializable. Its principal values are:

```
confidence_level = 0.95
minimum_independent_blocks = 4
minimum_block_frames = 16
strict_ediff_eV = 1e-6
warning_ediff_eV = 1e-4
strict_NVE_drift = 1e-3 eV/(atom ps)
hard_NVE_drift = 2.6e-2 eV/(atom ps)
catastrophic_energy_jump = 1 eV/atom/frame
catastrophic_force_norm = 100 eV/Angstrom
catastrophic_speed = 1000 Angstrom/ps
catastrophic_overlap = 0.35 Angstrom
strict_temperature_CI_half_width = max(5 K, 5% of mean)
strict_temperature_span_drift = max(5 K, 0.5 standard deviations)
strict_fixed_cell_relative_deviation = 1e-10
inactive_thermostat_energy_tolerance = 1e-8 eV
NVE_hard_reference_temperature = 300 K
NVE_hard_reference_time = 1 ps
```

These are policy values, not universal physical constants. Every result stores the policy signature and measured effect sizes so later versions can be replayed.

Public APIs

```
assess_trajectory_quality(...)
assess_vasp_trajectory_quality(...)
```

`read_vasp_frames` may attach a source-wide STAT0 verdict automatically when the full unstrided `vasprun.xml` trajectory is read. Subselected reads do not silently inherit a full-source verdict; their metadata records that a dedicated segment assessment is required.

Real-source acceptance

For the supplied 1,500-step Na-LTA source, the first implementation must:

- reconstruct ionic temperature from the complete **kinetic** channel;
- use 501 ionic degrees of freedom for 168 periodic atoms;
- report a mean near 320 K and a standard deviation near 14 K;
- resolve fixed-cell NVE from the ENS1 certificate;
- detect the approximately -0.205 eV/ps total-energy drift;
- classify the source as **degraded_quality**, not **unqualified**;
- continue to permit compatible structural and occupancy analyses;
- retain the misleading **SYSTEM** text only as a non-authoritative diagnostic.

Non-goals

STAT0 does not:

- infer a production interval;
- prove equilibrium or ergodicity;
- construct a canonical PMF;
- validate basin or saddle sampling;
- repair or detrend source data before downstream use;
- hide degraded evidence by replacing source channels with fitted values.